

PDF ANONIMIZADO

ID publico: case_0031

Paginas del documento: 10

Copia anonimizada para verificacion metodologica.

Se removieron portada editorial, titulo, autores, afiliaciones, correos, DOI, URLs y metadatos del PDF original.

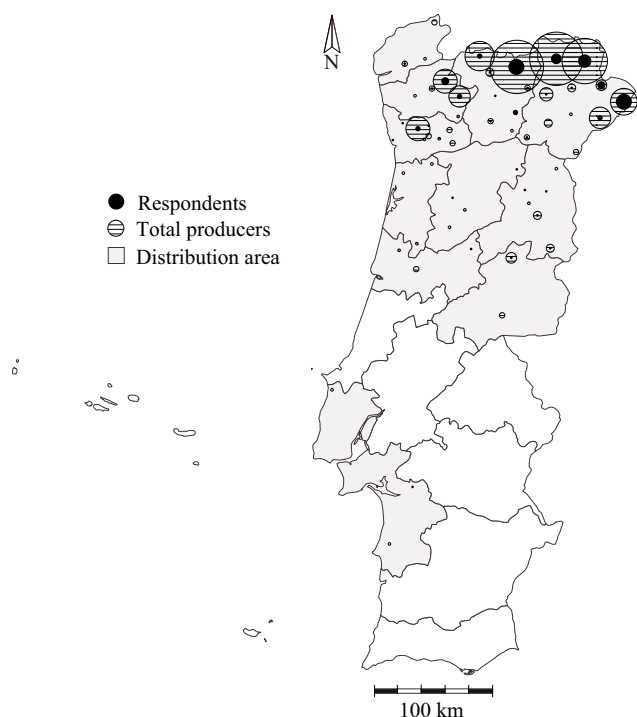
Las paginas restantes conservan el contenido util para revisar metodos, resultados, tablas y conclusiones.

Nota: la anonimización automatica prioriza reducir la identificacion directa del articulo. Los casos de una sola pagina o diseno no convencional deben revisarse manualmente antes de una publicacion abierta.

op appropriate and sustainable
 ísaro pig and other autochthonous
 l systems.

Material and Methods

The study was conducted between February and April 2017 as a disclosed identity mail survey to all producers registered in the National Bísaro Pig Producer Association (ANCSUB). Farms are concentrated in the northeast and dispersed to coastal and southern areas, in a lesser proportion, covering 11 of the 17 mainland Portugal districts (Figure 1). Notwithstanding, the majority of farms are located at a high altitude (over 400 m above sea level), with the topography ranging from inland plateaus to semi-mountainous and mountainous areas. The survey included all the production regions. At the time of the study, there were 6396 and 631 registered breeding sows and boars, respectively, from 215 producers. Only 194 producers had at least one registered animal on the farm, being considered as active, and, therefore, composed the study population. A



Size of circles represents proportion of respondents/total producers and shows the approximate location of farms.

into categories according to their types (LSU). Livestock units are European reference units, which facilitate the aggregation of livestock from various species and age as per convention. In pigs, breeding animals over 50 kg represent 0.5 LSU, piglets under 20 kg, 0.027 LSU, and all other pigs over three months old, 0.3 LSU. Smallholders (SH) include small farms with less than 15 LSU, whereas non-industrial medium-sized holdings (MS) comprise farms with 15 or more LSU. Industrial farms represent holdings with more than 260 LSU, with animals being exclusively reared on intensive systems.

Statistical analysis was conducted in JMP 7 (SAS Institute Inc., Cary, NC, 2007). Descriptive analysis was first performed to characterize the whole population trends. Categorical variables were presented as absolute and relative frequencies for both farm types, and the continuous variables were described as means $\pm$ SEM, median, range, or interquartile range. These variables were analyzed at the level of farm category, and group differences were identified using the chi-squared test for categorical variables. The Shapiro-Wilk test was used to assess normality for continuous variables of interest. For normally distributed data, farm differences were compared using ANOVA, whereas the Wilcoxon Rank test was used for non-normal distributed data. P-values (SH vs. MS) represent the probability of a farm category (independent variable) and the study parameter (dependent variable) being independent ($\alpha = 0.05$).

The estimated annual production and the annual meat production estimate were calculated, respectively, by the following equations: annual average number of animals sold per LSU $\times$ total LSU, and estimated annual production $\times$ carcass average weight. The average carcass weights considered for piglets, growers, and finishers were 6, 67, and 120 kg respectively (Costa, 2015; Fernandes et al., 2015). Average stocking density per breeding animal was estimated from the quotient between the values of indoor areas, obtained from individual responses, and the registered breeding stock in ANCSUB database.

Results

farm managers were 47 years old and had an average of eight-year experience rearing pigs. Furthermore, 84.5% of producers had finished secondary education (Table 1).

Figure 2 displays the estimated annual BP production, considering the average number of animals sold for slaughter each year per farm. Piglets for slaughter represented the vast majority of animals intended for slaughter per farm per year (91.1%). The estimated annual production was 834 tons of meat and 62692 slaughtered animals. Only 3.7% of the animals sold for slaughter are mature pigs intended for the cured meat market, representing an estimate of 275 tons of meat going for processing.

Almost all respondents (86.4%) declared rearing exclusively BP and only 6.9% affirmed crossing BP with other breeds. All farmers that responded using other swine breed claimed using commercial exotic breeds; Pietran had the largest number of answers ($n = 5$). On most MS farms (61.5%), pig production represented the main income for producers, whereas 21.2% of SH producers claimed to rear BP for recreational or traditional purposes. Within farm organization, significant differences between SH and MS farms were found in accounting ($P < 0.05$) and workforce ($P < 0.05$). Non-industrial medium-sized holdings farms were organized mainly as a one-person business with organized accounting (41.7%), while SH were organized as single producers with no business/enterprise structure (64.3%). Most of the SH workforce was composed of the producer and his family (78.6%); contrarily, most of the MS admitted having paid workers (66.7%). A fifth of the respondents had an associated meat-processing unit, which were predominantly classified as micro establishments.

Almost all BP producers (94.7%) claimed using feed other than commercial ration and 24.6% admitted not using any ration at all. Of other feed used, cereals (92.6%), vegetables (90.7%), and tubers (88.9%) were the main alternative or complement feed to commercial ration. Almost a fifth of the producers (22.2%) admitted using domestic food scraps to feed their animals (Figure 3). The majority of the respondents (79.3%) stated they differentiate feed between production groups. The existence of automated feeding and drinking systems depended on farm type ($P < 0.05$), and all MS farms had at least one automated system. Grazing did not differ between farm types, and 73.7% of all farmers claimed their animals have access to

	n (%)	54	20 (35.1)	14 (42.4)	6 (25.0)	0.528
	Mean±SEM		47.2±1.6	46.3±2.1	48.4±2.5	
	Median [Min, Max]		46 [24, 82]	45 [24, 82]	46 [32, 72]	
Experience (years)		58				0.279
	Mean±SEM		7.6±1.0	8.6±1.6	8.1±1.1	
	Median [1Q, 3Q]		5 [4, 10]	5 [1, 40]	7 [2, 25]	
Level of education		58				0.643
Primary education	n (%)		9 (15.5)	4 (12.1)	5 (20.0)	
Secondary education	n (%)		24 (41.4)	15 (45.5)	9 (36.0)	
Higher education	n (%)		25 (43.1)	14 (42.4)	11 (44.0)	
Farm profile						
Annual production for slaughter ³						
Piglets at weaning		57				<0.001
	Mean±SEM		323.7±46.3	158.3±20.4	535.3±86.1	
	Median [1Q, 3Q]		212 [103, 400]	150 [62, 251]	450 [256, 750]	
Growers - fresh meat		57				0.356
	Mean±SEM		18.5±15.9	3.3±1.4	37.9±36.3	
	Median [1Q, 3Q]		0 [0, 2]	0 [0, 3]	0 [0, 0]	
Finishers - cured meat		57				0.058
	Mean±SEM		13.0±4.9	2.9±0.7	25.9±10.8	
	Median [1Q, 3Q]		2 [0, 10]	2 [0, 5]	5 [0, 23]	
Annual production for sale ³						
Breeding animals		57				0.092
	Mean±SEM		6.6±3.6	2.2±1.6	12.3±8.0	
	Median [1Q, 3Q]		0 [0, 2]	0 [0, 0]	0 [0, 8]	
Fattening animals		57				0.929
	Mean±SEM		2.0±0.7	1.8±0.9	2.2±1.2	
	Median [1Q, 3Q]		0 [0, 0]	0 [0, 0]	0 [0, 0]	
Use of other swine breeds		59				1
Yes	n (%)		8 (13.6)	5 (15.2)	3 (11.5)	
No	n (%)		51 (86.4)	28 (84.8)	23 (88.5)	
Crossbreed		58				0.620
Yes	n (%)		4 (6.9)	3 (9.4)	1 (3.8)	
No	n (%)		54 (93.1)	29 (90.6)	25 (96.2)	
Purpose of rearing pigs		59				0.232
Primary income	n (%)		29 (49.2)	13 (39.4)	16 (61.5)	
Secondary income	n (%)		20 (33.9)	13 (39.4)	7 (26.9)	
Other	n (%)		10 (16.9)	7 (21.2)	3 (11.5)	
Accounting		52				0.035
Single producer	n (%)		25 (48.1)	18 (64.3)	7 (29.2)	
One-person business	n (%)		17 (32.7)	7 (25.0)	10 (41.7)	
Society/group	n (%)		10 (19.2)	3 (10.7)	7 (29.2)	
Workforce		49				0.006
Self-employed/family work	n (%)		29 (59.2)	22 (78.6)	7 (33.3)	
Some paid work	n (%)		16 (32.7)	5 (17.9)	11 (52.4)	
All paid work	n (%)		4 (8.2)	1 (3.6)	3 (14.3)	
Meat processing unit		59				0.187
Yes	n (%)		11 (18.6)	4 (12.1)	7 (26.9)	
No	n (%)		48 (81.4)	29 (87.9)	19 (73.1)	

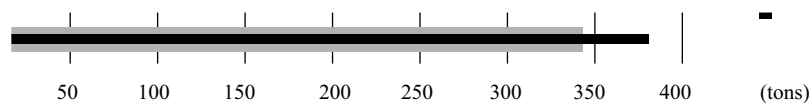


Figure 2 - Estimate of total Bísaro annual production (tons) based on the number of animals sold for slaughter (x1000).

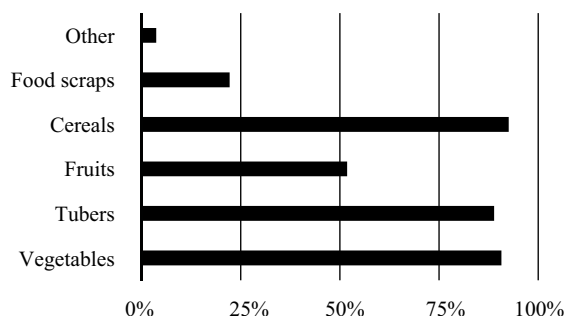


Figure 3 - Complement feed to commercial ration included in Bísaro production – percentage of respondents that use or include feed other than commercial ration in their animal diet, by class (n = 57).

grazing at part or full-time. Conversely, lactating sows (91.3%) and piglets (76.1%) were those mostly identified having diet supplementation (Figure 4).

Over half of the producers interviewed characterized their housing system as traditional (52.6%), having a small shed or traditional piggery with outdoor pens; however, almost another half (40.4%) described it as being fully outdoor, on a camping system. Few producers reported having forced ventilation in the animal building (18.2%), but MS farms were more likely to have this equipment ($P < 0.05$). Most of the producers admitted they separate animals by production group (84.5%); however, few seem to separate by gender (28.3%); furthermore, 53 days was the average age when animals were separated by gender. Maternity systems did not differ between farm categories ($P > 0.05$). On the other hand, maternity systems were dependent on housing systems ($P < 0.05$) with almost a third of producers allowing farrowing to happen in outdoor cabins (26.9%).

Most producers claimed their animals have access to outdoor areas (87.9%). The outdoor areas ranged from 56 m² to 40 ha (median = 1 ha; IQR: 0.2-2.8). The average outdoor area was larger in MS than in SH farms, although no significant differences were registered. In general, MS farms had also larger indoor areas compared with SH

Approximately one-third (34.5%) of the interviewed producers assumed they perform heat detection on their breeding sows. All of the producers admitted using natural mounting with own boar; few producers (12.5%) claimed using artificial insemination (AI); and only one (1.8%) reported breeding using boars from another Bísaro producer. Most of them introduced sows to boars at eight months of age (IQR: 6-15) and wean piglets at 35 days (IQR: 20-60). Few BP producers (15.8%) affirmed to voluntarily wait to breed sows again after weaning, which was more evident in SH ($P < 0.05$); three-quarters of these producers claimed to wait, on average, until the second standing heat after weaning. Moreover, fewer farmers claimed planning farrowing to happen in certain periods of the year (8.8%). The majority of producers (61.7%) assumed surgically castrating males, usually just before weaning (median: 29 days; IQR: 4-75). On average, BP producers that responded to the survey believe having two litters per sow annually and eight in their productive lifetime.

Discussion

The data collected in this study came from almost a third of the population of producers and, therefore, represent a good overview of the practices used in Bísaro production nowadays. Categorization of BP farms based on hierarchical clusters resulted in artificial and disparate groups regarding the number of farms included. Despite the difficulties in grouping a large number of producers by their production systems, categorization based on LSU constituted a better representation of the BP reality. Smallholders belonged to single producers and had family-based work; automation was absent, and production regime was predominantly less strict. Non-industrial medium-sized holdings farms were commercial-based farms, with larger production, greater farm areas, and were technologically more advanced.

Younger, less experienced, and more educated farmers resumed the demography of producers. More than half of

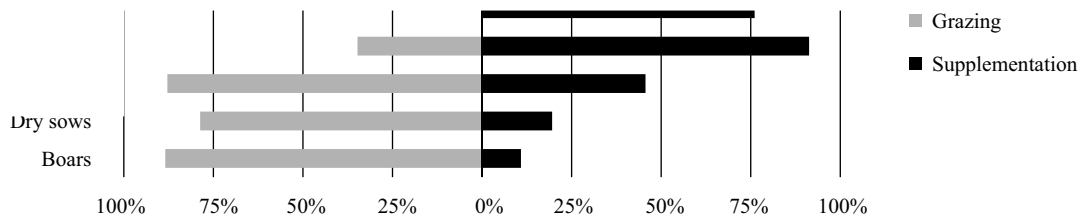


Figure 4 - Grazing and diet supplementation in Bísaro production – percentage of producers that contemplate grazing and supplementation in their feeding regimes, by production group (n = 57).

pig production, but are contrary to older descriptions from a Bísaro technical report (Outor-Monteiro et al., 2005). Experience in rearing pigs was, however, significantly lower when compared with other survey-based studies regarding pig production in Asia and Oceania (Alawneh et al., 2014; Leslie et al., 2015; Schembri et al., 2015). Controversially, most of the respondents held a higher education degree, which could be partly explained by the EU financial aid for young farmers to establish new holdings of the last decades. This phenomenon is truly decisive given the higher European funding for endangered breeds, like the BP, compared with current industrial hybrids. It also explains why so many of these smallholders (39.4%) can live primarily sustained by this activity. Surprisingly, some smallholders also stated having some paid work (17.9%), mainly seasonal-related with animal handling and agricultural labour spikes across the year.

Although a previous field study (Carvalho, 2000) described smoked-cured meat as the majority of BP production, our results showed a considerable higher number of animals sold for roast piglet, usually after weaning (10-12 kg BW) compared with animals slaughtered for fresh or cured meat. In fact, our estimate of BP meat production per year follows the same trend, in which piglets were the most representative carcass weight produced compared with other classes for slaughter. Furthermore, and contrary to existing technical reports from ANCSUB (2007), a third of the respondents reported to exclusively produce and sell piglets for slaughter; these farms could be categorized as “breeding only”. The remaining producers admitted selling both groups of animals, falling into a “farrow-to-finish” category.

Regarding animal husbandry, practices were dissonant amongst feeding, housing, and reproduction. Traditional feeding, in which commercial ration is completed with

(Outor-Monteiro et al., 2005; Fernandes et al., 2015), feeding substantially consisted of own-farm crops: cereals, harvested vegetables, tubers, and fruit. Most of the animals had access to pasture at some point, but boars and pregnant and dry sows mostly benefit from this practice. Inversely, the majority of lactating sows and piglets did not benefit from grazing, but rather were supplemented or had differentiated feed to fulfil their needs. On the other hand, housing seems to have changed over the last two decades. Carvalho (2000) reported all animals were kept in permanent stabling. Conversely, our study demonstrated that almost half of the producers have now their animals in permanent outdoor camping systems. Moreover, nearly all the producers have stated their animals have access to outside areas. Notwithstanding, there are still industrial and traditional piggeries used for housing the animals for great part of their production life. These systems had $16.8 \pm 2.3 \text{ m}^2$ per breeding stock on average. This stocking density is well above the recommended stock density by EU directive (Council directive 2008/120/EC, 2009) ($1.64\text{--}2.25 \text{ m}^2$ per gilts and sows, respectively, when kept in groups of 6-39 animals), but the large variance registered could indicate some farms may be compromising stocking densities. In fact, two farms had indoor areas per breeding animal below 2 m^2 . Despite the contradictory reports from ANCSUB (2007), a third of producers claimed using farrowing crates.

Reproduction practices are generally poorer when compared with industrial pig production trends. Heat detection is not commonly used, suggesting that often the stud boar is kept in heterosexual groups. The use of artificial insemination is still incipient, and only one producer claimed to use boars from other farms; these two reproductive methods are important genetic diversity promoters and fundamental to a small population with a

for alternative rearing systems (9-
., 2013; Karolyi et al., 2016).
male piglets is still a common
practice in the BP production system, even though

piglet were not being subjected to the procedure. This
tendency antagonizes with the rest of Portuguese industrial
pork production, where most of males are kept entire
(Thun et al., 2006; Fredriksen et al., 2009). Most of these

Table 2 - Descriptive statistics of feeding practices and housing conditions by type of farm

	Units ¹	R	Total	Smallholders	Medium-sized farms	P-value ²
Feeding						
Proportion of ration used		57				0.109
None	n (%)		14 (24.6)	9 (29.0)	5 (19.2)	
<50	n (%)		20 (35.1)	13 (41.9)	7 (26.9)	
50-75	n (%)		10 (17.5)	6 (19.4)	4 (15.4)	
>75	n (%)		10 (17.5)	3 (9.7)	7 (26.9)	
100	n (%)		3 (5.3)	0 (0.0)	3 (11.5)	
Differentiated feeding		58				0.518
Yes	n (%)		46 (79.3)	24 (75.0)	22 (84.6)	
No	n (%)		12 (20.7)	8 (25.0)	4 (15.4)	
Automated feed/drinking systems		58				<0.001
None	n (%)		13 (22.4)	13 (40.6)	0 (0.0)	
Automated water troughs only	n (%)		34 (58.6)	17 (53.1)	17 (65.4)	
Automated water troughs and feeders			11 (19.0)	2 (6.3)	9 (34.6)	
Grazing		57				0.765
Yes	n (%)		42 (73.7)	22 (71.0)	20 (76.9)	
No	n (%)		15 (26.3)	9 (29.0)	6 (23.1)	
Housing						
Housing system		57				0.399
Industrial	n (%)		4 (7.0)	2 (6.5)	2 (7.7)	
Traditional	n (%)		30 (52.6)	14 (45.2)	16 (61.5)	
Camping	n (%)		23 (40.4)	15 (48.4)	8 (30.8)	
Forced ventilation		55				0.035
Yes	n (%)		10 (18.2)	2 (6.9)	8 (30.8)	
No	n (%)		45 (81.8)	27 (93.1)	18 (69.2)	
Separation by production group		58				0.495
Yes	n (%)		49 (84.5)	26 (81.3)	23 (88.5)	
No	n (%)		9 (15.5)	6 (18.8)	3 (11.5)	
Separation by gender		46				0.205
Yes	n (%)		13 (28.3)	5 (20.0)	8 (38.1)	
No	n (%)		33 (71.7)	20 (80.0)	13 (61.9)	
Maternity system		52				0.298
Farrowing crates with thermal insulation	n (%)		18 (34.6)	8 (26.7)	10 (45.5)	
No farrowing crates/thermal insulation	n (%)		20 (38.5)	12 (40.0)	8 (36.4)	
Outdoor cabins	n (%)		14 (26.9)	10 (33.3)	4 (18.2)	
Access to outside areas		58				0.225
Yes	n (%)		51 (87.9)	30 (93.8)	21 (80.8)	
No	n (%)		7 (12.1)	2 (6.3)	5 (19.2)	
Total outdoor area (ha)		51				0.099
	Mean±SEM		3.1±0.9	1.5±0.3	5.3±1.9	
	Median [1Q, 3Q]		1 [0.2, 2.8]	1 [0.2, 1.8]	2 [0.1, 5]	
Total indoor area (m ²)		52				<0.001
	Mean±SEM		252.4±32.2	129.1±16.3	407.8±55.2	
	Median [1Q, 3Q]		175 [100, 400]	100 [69, 193]	400 [180, 640]	

It was not possible to determine re being performed by a qualified mer. This should be highlighted on any subsequent study as EU regulations define that surgical castration on animals older than seven days should only be done under anaesthesia and prolonged analgesia by a veterinarian (Council directive 2008/120/EC, 2009). Although the UE directive 2008/120/EC restricts female spaying, undocumented reports from producers and workers also relate spaying gilts as a common practice, based on the assumption that cyclic ovarian function in sows will affect meat quality and, therefore, affect the acceptance of the final smoked-cured products. However, in this survey, only one producer admitted keeping this practice.

domestic food scraps to their animals in SH (25%) and, to a lesser extent scale, on MS farms (17.3%). These figures are higher than results from previous studies in other countries, where 1.5% (Relun et al., 2015) and 8% (Ribbens et al., 2008) of farmers reported feeding table scraps to their pigs. In sum, a significant number of irregular practices were reported by producers in both types of production system. This figure may have been even higher if a face-to-face interview had been conducted instead. Education should always be prioritized to mitigate this issue, altogether with strategic modernization of the reproductive management in BP production system.

Collected data evidenced some important differences between SH and MS farms; yet, production traits and

Table 3 - Descriptive statistics of reproduction practices by type of farm

	Units ¹	R	Total	Smallholders	Medium-sized farms	P-value ²
Reproduction						
Heat detection		55				0.777
Yes	n (%)		19 (34.5)	11 (37.9)	8 (30.8)	
No	n (%)		36 (65.5)	18 (62.1)	18 (69.2)	
Reproduction method		56				
Natural with own boar	n (%)		56 (100.0)	30 (100.0)	26 (100.0)	1
Natural with external boar	n (%)		1 (1.8)	1 (3.3)	0 (0.0)	1
Artificial insemination	n (%)		7 (12.5)	5 (16.7)	2 (7.7)	0.431
Age at first service (months)		53				0.756
	Mean±SEM		8.5±0.2	8.6±0.3	8.5±0.4	
	Median [1Q, 3Q]		8 [8, 9]	8 [8, 10]	8 [7, 9]	
Planned farrowing		57				0.056
Yes	n (%)		5 (8.8)	5 (16.1)	0 (0.0)	
No	n (%)		52 (91.2)	26 (83.9)	26 (100.0)	
Age at weaning (days)		56				0.155
	Mean±SEM		38.0±1.2	39.6±1.7	35.9±1.7	
	Median [1Q, 3Q]		35 [30, 45]	38.5 [30, 45]	35 [30, 40]	
Voluntary waiting period		57				0.031
Yes	n (%)		9 (15.8)	8 (25.8)	1 (3.8)	
No	n (%)		48 (84.2)	23 (74.2)	25 (96.2)	
Surgical castration		47				0.426
Males only	n (%)		28 (59.6)	14 (56.0)	14 (63.6)	
Males and females	n (%)		1 (2.1)	0 (0.0)	1 (4.5)	
None	n (%)		18 (38.3)	11 (44.0)	7 (31.8)	
Age at castration (days)		24				0.037
	Mean±SEM		27.5±3.9	36.2±6.2	18.8±3.3	
	Median [1Q, 3Q]		29 [9, 35]	33 [19, 56]	19 [7, 30]	
Litters per sow per year		57				0.159
	Mean±SEM		2.0±0	1.9±0.1	2.1±0.1	
	Median [1Q, 3Q]		2 [2, 2]	2 [2, 2]	2 [2, 2]	
Litters per sow per lifetime		43				0.781
	Mean±SEM		7.8±0.5	7.7±0.8	8.0±0.5	
	Median [Min, Max]		8 [2, 16]	7 [2, 16]	8 [5, 14]	

es an important weakness within
Nevertheless, animal husbandry
proved from previous reports,
especially those related to housing. A significant proportion
of Bísaro farms operate now on free-range systems, and
most of them allow access to pasture. This important trait
could be decisive when there is a growing market for
informed consumers seeking for premium animal-friendly
products.

Conclusions

The present study characterizes the Bísaro production
system and increases our knowledge of traditional
management practices of native pig breeds reared in
alternative systems. It also supports future decision-
making to improve the breed sustainability, considering the
constraints raised by traditional practices now described.

Some of the practices acquired in this study reveal
noncompliance with European Union regulations and must
be revised, while others demonstrate variability within
the production system and, therefore, should be discussed
and delineated to enhance consistency of Bísaro products.
Even though Bísaro farms can be typified into smallholders
and medium-sized farms based on production, structure,
automation and equipment, indoor areas, and reproductive
management practices.

Bísaro conservation and sustained development should
be prioritized to maximize animal production yields
and profit of the producer, meeting the expectations of
traditional cured meat production markets.

Acknowledgments

The authors wish to thank all producers, for their
contribution, and ANCSUB and RuralBit, for their
technical support. This study was sponsored by the
Portuguese Science and Technology Foundation
(FCT) under Projects UID/CVT/00772/2013 and UID/
CVT/00772/2016. Gustavo Paixão holds a PhD grant from
the Animal Science Doctoral Program (AniSci), operation
number NORTE-08-5369-FSE-000040, co-funded by the
European Social Fund (FSE) and through National funds
NORTE 2020 and Regional Operational Programme of

Araújo, J. P.; Amotim, I.; Santos Silva, J.; Pires, P. and Cerqueira, J.
2016. Outdoor housing systems for Bísaro pig breed with a hoop
barn: some effects on welfare. p.87-91. In: Food futures: ethics,
science and culture. Wageningen Academic Publisher, Porto.

Carvalho, M. 2000. Nutrition and production systems of the "Bísaro"
pig in the north-east region of Portugal. Options Méditerranéennes.
Série A. Seminaires Méditerranéens. p.123-128.

Costa, H. M. G. 2015. Estudo prévio sobre a carcaça de suínos da
Raça Bísara, de peso vivo de 10 a 65 kg. Dissertação (M.Sc.).
Instituto Politécnico de Bragança, Portugal.

Fernandes, A. J.; Fernandes, A.; Teixeira, A.; Leite, A.; Gonçalves,
A.; Amorim, A.; Oliveira, A.; Alves, C.; Pereira, E.; Paulos, K.;
Pereira, L. and Rodrigues, S. 2015. Porco Bísaro - Qualidade da
carcaça e da carne. Quinta do Bísaro, Bragança, Portugal.

Fernandes, S. D.; Malovrh, S. and Kovac, M. 2010. Study of genetic
diversity of bisaro pigs breed by pedigree analysis. *Seria Zootehnie*
- Universitatea de Stiinte Agricole si Medicina Veterinara Ion
Ionescu de la Brad 178:326-330.

Karolyi, D.; Luković, Z.; Škorput, D.; Mahnet, Ž.; Klišanić, V.;
Vnučec, I.; Salajpal, K. and Bošnjak, A. 2016. Morphological and
reproductive traits of Turopolje pig breeding sows. p.47. In: 9th
International Symposium on the Mediterranean Pig. Portalegre,
Portugal.

-92.

veia, S.; Evans, G.; Zhang, Y.; Cardoso, w, G. and Rangel-Figueiredo, T. 2003. s for initial genetic characterization of two Portuguese swine breeds. *Archivos de Zootecnia* 52:255-264.

Switzerland.

Santos Silva, J. and Tirapicos Nunes, J. L. 2013. Inventory and characterization of traditional Mediterranean pig production systems. Advantages and constraints towards its development. *Acta Agriculturae Slovenica Supplement* 4:61-67.

Shih, T. and Fan, X. 2007. Response rates and mode preferences in web-mail mixed-mode surveys: a meta-analysis. *International Journal of Internet Science* 2:59-82.

Thun, R.; Gajewski, Z. and Janett, F. 2006. Castration in male pigs: techniques and animal welfare issues. *Journal of Physiology and Pharmacology* 57(Suppl 8):189-94.

Santos e Silva, J.; Bernardo, A. and Pires da Costa, J. 2000a. Genetic characterization and inventory of the Bisaro pig through visible effect genes. Their utilization in the genotypic comparison between populations and in the establishing of a nucleus for in vivo genetic conservation. *Options Méditerranéennes. Série A, Séminaires Méditerranéens* 41:39-51.